

Bridge Solver X-ray

An instrumented version of macroxue’s bridge-solver with enhanced debugging and tracing capabilities.

Purpose

This repository maintains an instrumented (“X-ray”) version of the bridge double dummy solver for:

- **Debugging:** Trace solver execution with detailed logging of search states, move ordering, alpha-beta cutoffs, and transposition table operations
- **Validation:** Verify that modified or reimplemented solvers produce identical results to the reference implementation
- **Understanding:** Study the solver’s decision-making process for educational purposes

Repository Structure

- **Root:** Original upstream solver files (unmodified)
- **xray/:** Instrumented solver, tests, and build files

Building

```
cd xray
make
```

This builds `solver-xray` from `bridge-solver.cc`.

Running Tests

```
cd xray
make test
```

Runs the solver against 100 test deals and compares results to golden reference output.

X-ray Debugging Options

The instrumented solver supports additional command-line options:

Option	Description
-X N	Enable X-ray tracing for first N SearchWithCache calls
-P	Disable pruning (fast/slow tricks optimization)
-T	Disable transposition table
-R	Disable min_relevant_ranks optimization

Example: Trace first 10 search calls

```
cd xray
./solver-xray -f test_deals/deal.1 -X 10
```

Logging Categories

When X-ray tracing is enabled, the solver outputs detailed logs:

- **XRAY**: Entry point with hands and play history
- **FAST_TRICKS** / **SLOW_TRICKS**: Pruning calculations
- **MOVE_ORDER**: Card ordering before/after evaluation
- **SCORE**: Individual move scores during search
- **CUTOFF**: Alpha-beta cutoff events
- **EQUIV_V2**: Card equivalence checking

Upstream

This repository tracks macroxue/bridge-solver.

Original Documentation

Below is the original README from the upstream solver.

Bridge double dummy solver

This is a fairly simple and yet effective double dummy solver for the card game of bridge. It's terminal based.

Solve a random deal

```
cd xray
./solver-xray -r
```

The output looks like below.

```

                KJT987  K5  7  AQJ8
      3  J9764  Q642  KT2                Q64  QT8  KJ953  94
                A52  A32  AT8  7653
N 13 13  0  0  0.01 s 10.2 M
S 13 13  0  0  0.02 s 10.2 M
H  7  7  6  5  0.21 s 12.9 M
D  6  6  6  6  0.34 s 13.2 M
C 13 13  0  0  0.34 s 13.2 M
```

Each line after the deal shows the strain to play, the number of tricks when South/North/West/East declares respectively, the cumulative time and the peak memory usage.

Solve a deal in a file

```
cd xray
./solver-xray -f FILE
```

The format of the deal in the file is like below.

```

                                KQ3 - T832 AJ9765
72 AJ972 AQ7 KQ2                                T96 K83 654 T843
                                AJ854 QT654 KJ9 -
D
W
```

The first line is North. The second line has both West and East. The third line is South. The fourth line specifies the strain to play. The fifth line is the leading seat. If the leading seat is not given, the deal is solved for all four leading seats. If the strain to play is also not given, the deal is solved for all five strains.

Interactive play

```
cd xray
./solver-xray -r -p
```

or

```
./solver-xray -f FILE -p
```

The solver automatically determines the contract. If nobody can make any contract, the hand is skipped. For each turn, the solver evaluates each of the player's card and shows the result of the contract if the card is played and the rest is played by everyone optimally. A sign is shown next to each card with the following meanings. | Sign | Meaning | |——|———| | = | The contract makes. | | + | The contract gets an overtrick. | | - | The contract is set by a trick. | | (+N) | The contract gets N overtricks. | | (-N) | The contract is set by N tricks. |

Performance

The original solver benchmarks are available in the upstream repository.